WHZ Westsächsische Hochschule Zwickau
University of Applied Sciences

WESTSÄCHSISCHE HOCHSCHULE ZWICKAU

University of Applied Sciences

Fakultät Physikalische Technik/Informatik

Practical Training Report

Environmental and Process Monitoring

Experiment 3

Nuclear Magnetic Resonance (NMR) Spectroscopy

Structural Identification and Quantitative Analysis

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1 Experimental Objectives

This practical training's main objective was to use NMR spectroscopy for both structural identification and quantitative analysis. The following was the definition of the particular experimental tasks:

1. To construct a calibration function based on the integration of a diagnostic resonance signal and utilize it to determine the concentration of glyphosate in a sample of unknown content.
2. To characterize the primary component of an unknown sample by interpreting its 1D (^{1}H, ^{13}C, DEPT) and 2D NMR spectral data, ultimately identifying the chemical structure of the compound.

2 Theory

2.1 NMR Spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy studies the magnetic properties of specific atomic nuclei, most notably the hydrogen nucleus (proton) and the carbon-13 isotope of carbon. NMR analysis of a molecule can reveal differences in the magnetic properties of these nuclei, allowing the positions of the atoms within the molecule to be determined. This technique allows you to identify the number of distinct chemical environments present, determine which atoms are close together, and, in most cases, estimate how many atoms occupy each environment(Kemp, 1991).

2.2 Basic principle of NMR

- **The Spinning Nucleus:** The proton, the nucleus of hydrogen, acts like a tiny spinning bar magnet. The combination of intrinsic mechanical spin and electric charge causes this magnetic behavior. This spinning motion creates a magnetic field, just like any charged particle in motion.
- **Impact of an External Magnetic Field:** The proton reacts to an external magnetic field like a compass needle, trying to align itself with the applied field. The proton can only take on two possible orientations due to quantum mechanical constraints: one that is aligned parallel to the field, signifying a lower energy state, and the other that is aligned antiparallel to the field, signifying a higher energy state (Kemp, 1991).

When a proton is aligned with a magnetic field, it has less energy. It has the ability to absorb energy and move to a higher-energy position that is opposite the magnetic field. After a brief

period, the proton releases this energy and returns to its lower-energy state. If radiofrequency energy at the appropriate frequency is applied, the proton will absorb it. This absorption occurs only when the radiofrequency corresponds to the proton's spinning (precession) frequency. This condition is referred to as resonance, and the technique is known as nuclear magnetic resonance.

A basic NMR experiment involves placing the protons in a molecule in a strong magnetic field, causing them to spin at different frequencies. Radiofrequency energy is then applied, causing the protons to transition from a lower-energy to a higher-energy state. The energy absorbed during this process is measured and shown as an NMR spectrum (Kemp, 1991).

2.3 Magnetic Nuclei and Spin Quantum Number

Not all nuclei show the NMR effect; only those with a spin quantum number $I > 0$ are magnetic. The spin quantum number I depends on the mass number and atomic number of the nucleus, as summarized below:

Table 1: Spin Quantum Numbers (I) for Selected Nuclei

Nucleus	**Mass Number**	**Spin Quantum Number (I)**
^{1}H (Proton)	1 (Odd)	1/2
^{12}C	12 (Even)	0
^{13}C	13 (Odd)	1/2
^{14}N	14 (Even)	1
^{15}N	15 (Odd)	1/2
^{16}O	16 (Even)	0
^{2}H (Deuterium)	2 (Even)	1
^{19}F	19 (Odd)	1/2
^{31}P	31 (Odd)	1/2

For example, the proton (^{1}H) has $I = 1/2$, while ^{12}C and ^{16}O have $I = 0$ and are therefore non-magnetic. If ^{12}C and ^{16}O were magnetic, NMR spectra of organic molecules would be far more complex.

Other commonly studied magnetic nuclei include ^{1}H, ^{13}C, ^{14}N, ^{15}N, ^{17}O, ^{19}F, and ^{31}P. Both deuterium (^{2}H) and nitrogen-14 (^{14}N) have $I = 1$, and this has important consequences, as will be discussed in later sections (e.g., heteronuclear coupling).

When placed in an external magnetic field, a magnetic nucleus can align in different orientations relative to the field. The number of possible orientations is given by $(2I + 1)$. Nuclei with $I = 1/2$ (e.g., ^{1}H, ^{13}C, ^{19}F) have two possible orientations, while nuclei with $I = 1$ (e.g., ^{2}H, ^{14}N) have three orientations. Nuclei with $I = 1$ not only have magnetic dipoles, but also electric quadrupoles, which allow them to interact with both magnetic and electric field gradients. The relative effects of these interactions depend on the magnitudes of the nuclear magnetic and

quadrupole moments (Kemp, 1991).

In an applied magnetic field, magnetic nuclei like protons precess at a frequency ν, which is proportional to the strength of the applied field. The exact frequency is given by:

$$\nu = \frac{\gamma}{2\pi} B_0$$

where B_0 is the strength of the external magnetic field experienced by the nucleus, and γ is the magnetogyric ratio, defined as the ratio between the nuclear magnetic moment μ and the nuclear angular momentum I.

2.4 Working of NMR spectrometer

A nuclear magnetic resonance (NMR) spectrometer is an analytical equipment that examines the magnetic characteristics of atomic nuclei. It is based on the interaction of externally applied radiofrequency (RF) radiation with nuclei with non-zero spin in a high magnetic field. A schematic diagram is shown below:

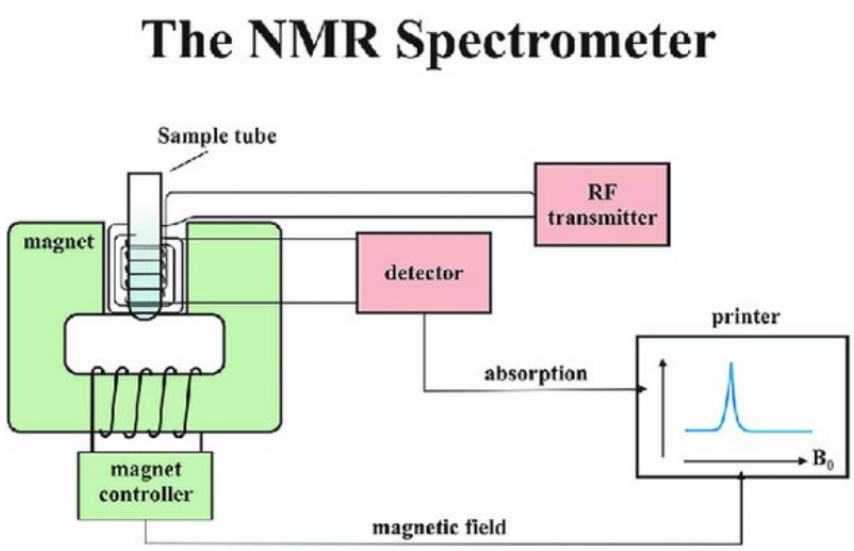

Figure 1: Schematic diagram of NMR Spectrometer (Smith and Strange, 1996)

Nuclei with spin ($I > 0$), such as protons (^{1}H) or carbon-13 (^{13}C), initially align with or against the strong external magnetic field generated by the superconducting magnet (B_0). The sample is placed inside the probe and then exposed to a pulse of radiofrequency (RF) radiation from the RF transmitter at the nucleus's Larmor frequency. This pulse causes nuclei to shift from the

lower-energy aligned state to the higher-energy opposed state. As the nuclei relax back to a lower-energy state, they emit RF signals that the probe's receiver coil detects. The console and computer process these signals using Fourier Transform to generate the NMR spectrum. The resulting spectrum contains detailed information about the chemical environment and arrangement of nuclei in the sample(Smith and Strange, 1996).

3 Experimental Procedure

A SPINSOLVE 80 ULTRA Multi-X benchtop NMR spectrometer from Magritek, equipped for analyzing ^{7}Li, ^{13}C, ^{23}Na, ^{29}Si, and ^{31}P, and including an autosampler, was used for the experiments. Data evaluation was performed using the SPINSOLVE® and MNova® software packages Magritek Ltd., 2023.

3.1 Quantification of Glyphosate Sample (qNMR Experiments Part I)

1. Perform a series of measurements to create a calibration function for quantifying an unknown concentration of a glyphosate sample:

   I. Set up a series of ^{31}P NMR experiments with the given parameters using the provided NMR samples at specified concentrations.

   II. Carry out the ^{31}P NMR experiments using the autosampler.

   III. Analyze and evaluate the recorded spectra using the software packages, following the given data processing protocol.

   IV. Generate a calibration function considering the integrals of a suitable resonance signal (Data evaluation).

3.2 Quantification of an Unknown Glyphosate Sample (qNMR Experiments Part II)

1. Perform an NMR experiment to quantify an unknown concentration of a glyphosate sample:

   I. Set up a ^{31}P NMR experiment with the given parameters using the provided NMR sample with an unknown concentration.

   II. Carry out the ^{31}P NMR experiment using the autosampler.

   III. Analyze and evaluate the recorded spectrum using the software packages, following the given data processing protocol.

   IV. Determine the glyphosate concentration using the calibration function of a suitable resonance signal (Data evaluation).

3.3 Analysis of an Unknown Sample (Same as IR Experiment)

1. Analyze an unknown sample:

   I. Set up a series of experiments (^{1}H, ^{13}C, DEPT-45, DEPT-90, DEPT-135, and ^{1}H-^{13}C HMQC) with the given parameters using the provided NMR sample dissolved in $CDCl_3$.

   II. Carry out the NMR experiments using the autosampler.

   III. Analyze and evaluate the recorded spectra using the software packages, following the given data processing protocol.

   IV. Determine the unknown substance based on the recorded spectra (Data evaluation).

3.4 Data Evaluation

1. Export the recorded spectra in CSV format and save them.

2. Write an experimental report presenting the recorded spectra and addressing the subtasks mentioned above.

3. Determine the unknown concentration of the glyphosate sample using the calibration function constructed earlier.

4. Identify the unknown sample from the recorded NMR experiments and compare the spectra with the NMR spectra stored in the SDBS database (https://sdbs.db.aist.go.jp).

3.5 Measurement of Data and evaluation

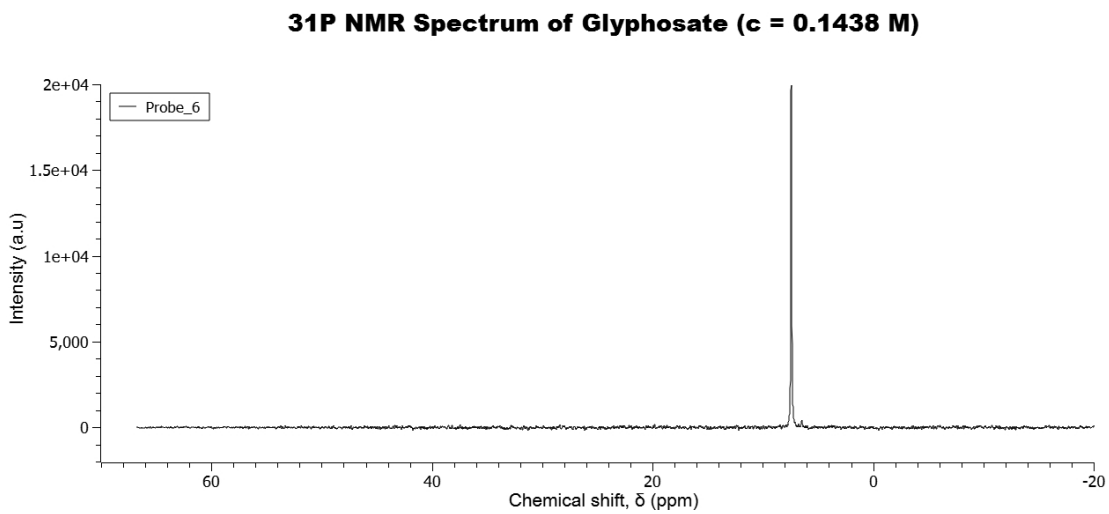

Figure 2: NMR Spectroscopy of 31P decoupled Glyphosate Probe 6 (c=0.1438 M)

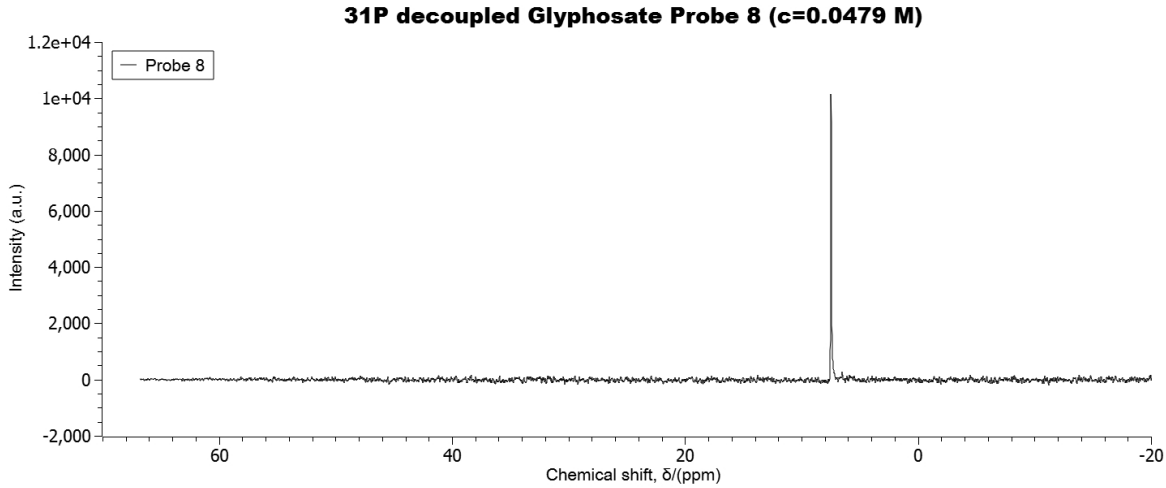

Figure 3: NMR Spectroscopy of 31P decoupled Glyphosate Probe 8 (c=0.0479 M)

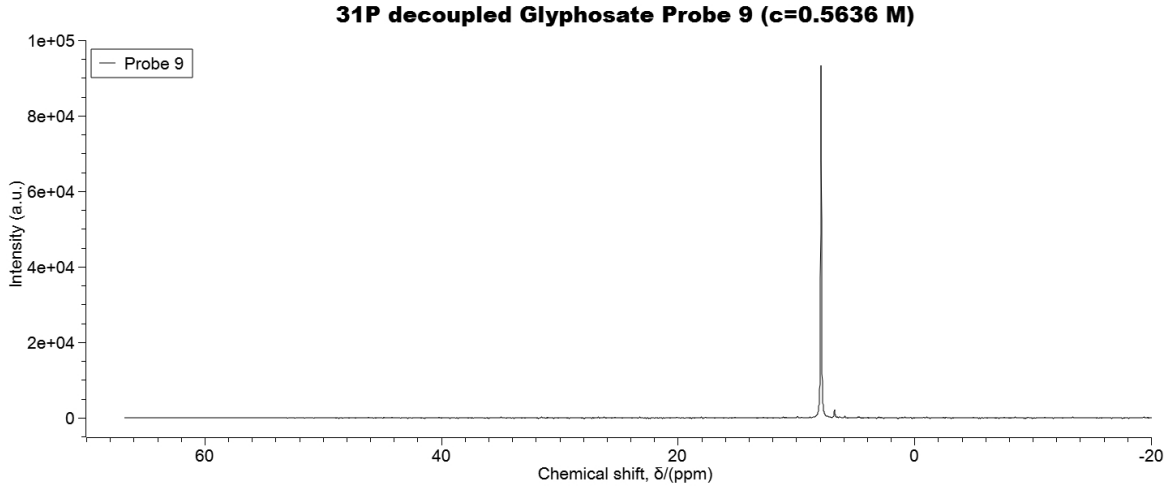

Figure 4: NMR Spectroscopy of 31P decoupled Glyphosate Probe 9 (c=0.5636 M)

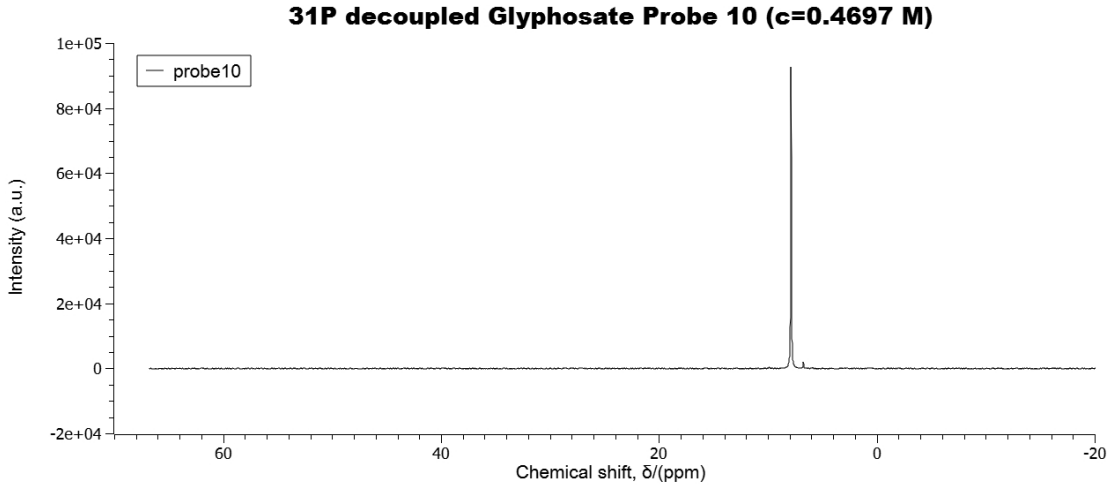

Figure 5: NMR Spectroscopy of 31P decoupled Glyphosate Probe 10 (c=0.4697 M)

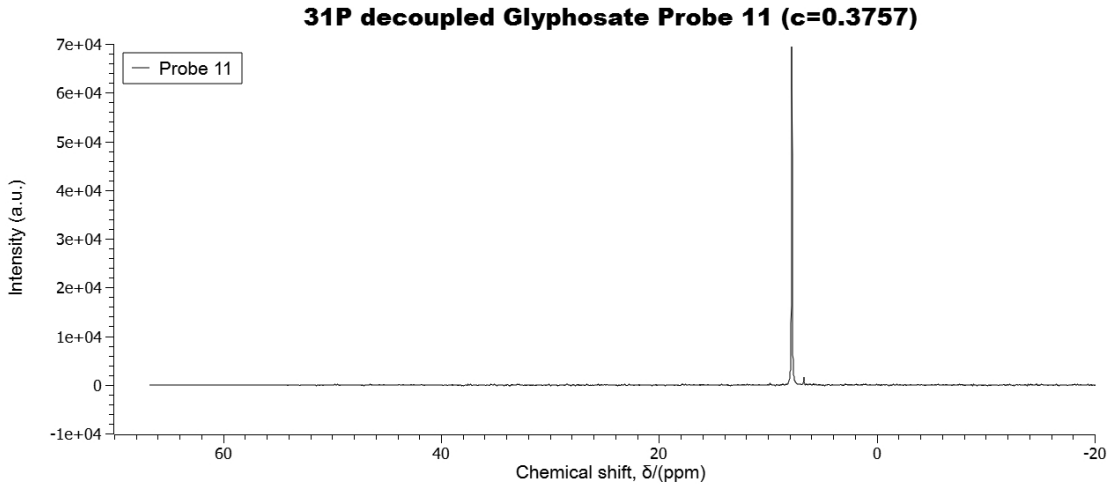

Figure 6: NMR Spectroscopy of 31P decoupled Glyphosate Probe 11 (c=0.3757)

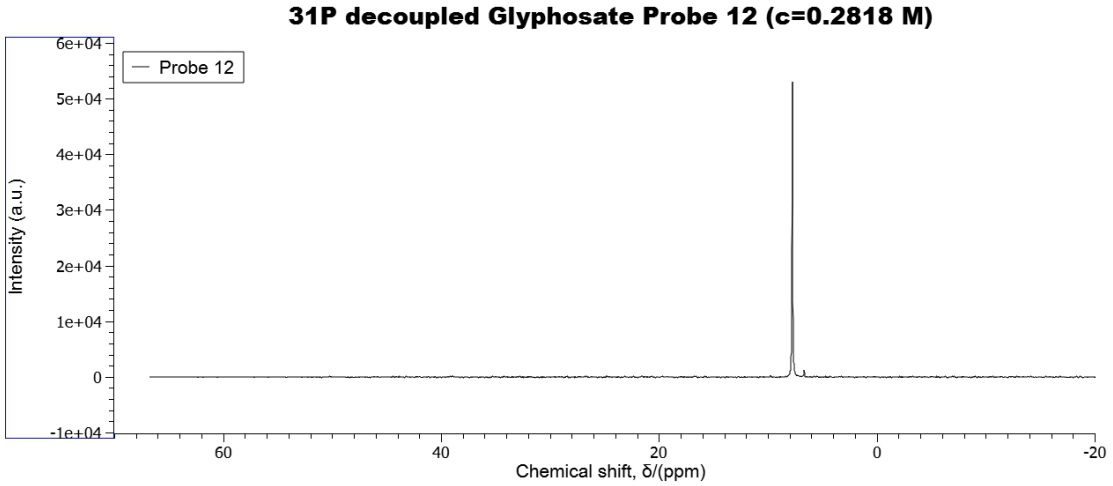

Figure 7: NMR Spectroscopy of 31P decoupled Glyphosate Probe 12 (c=0.2818 M)

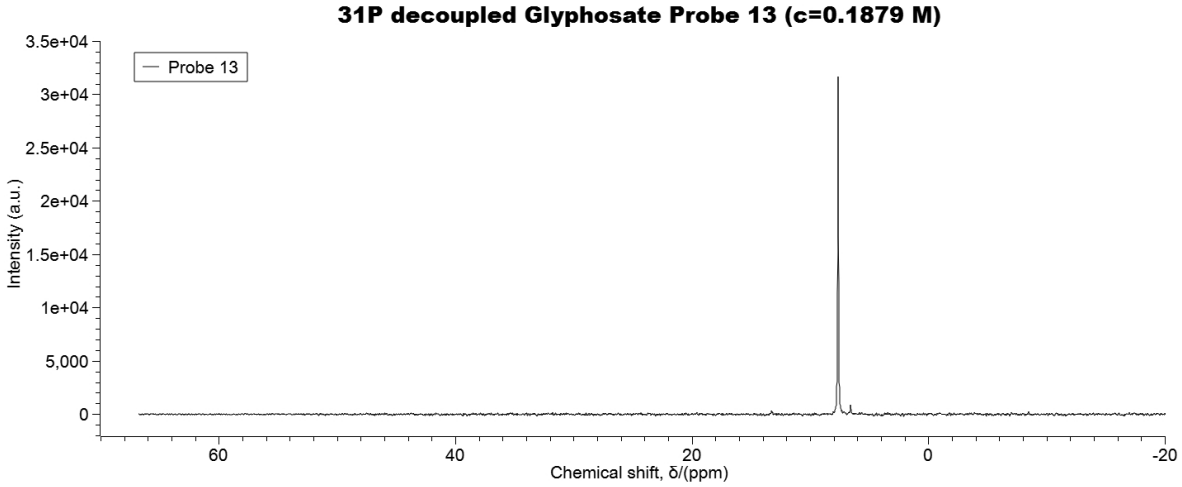

Figure 8: NMR Spectroscopy of 31P decoupled Glyphosate Probe 13 (c=0.1879 M)

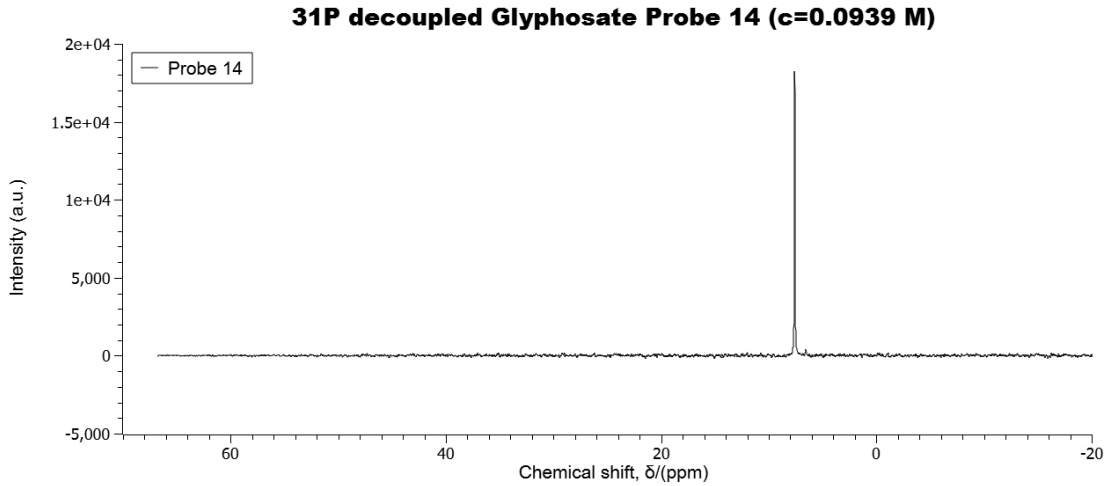

Figure 9: NMR Spectroscopy of 31P decoupled Glyphosate Probe 14 (c=0.0939 M)

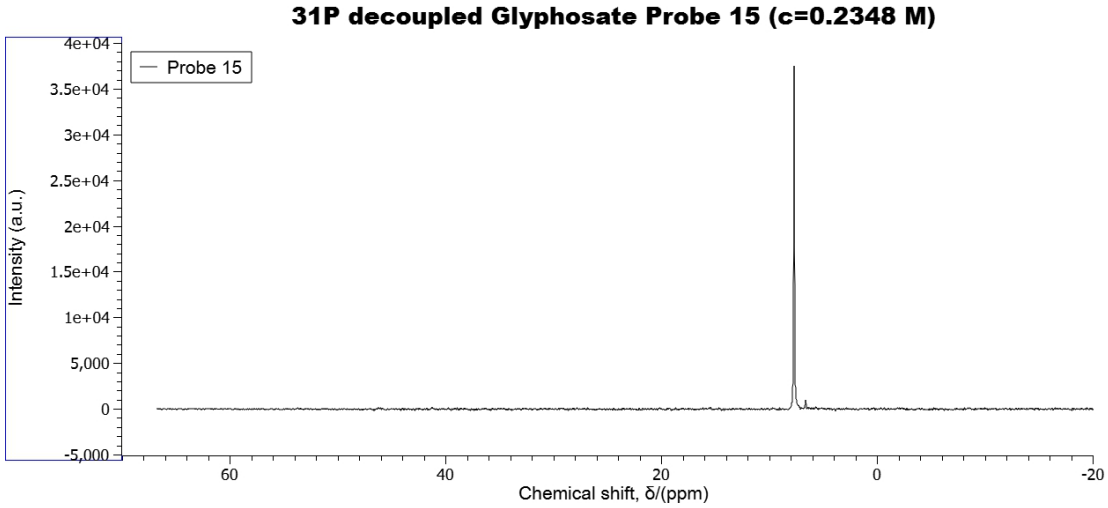

Figure 10: NMR Spectroscopy of 31P decoupled Glyphosate Probe 15 (c=0.2348 M)

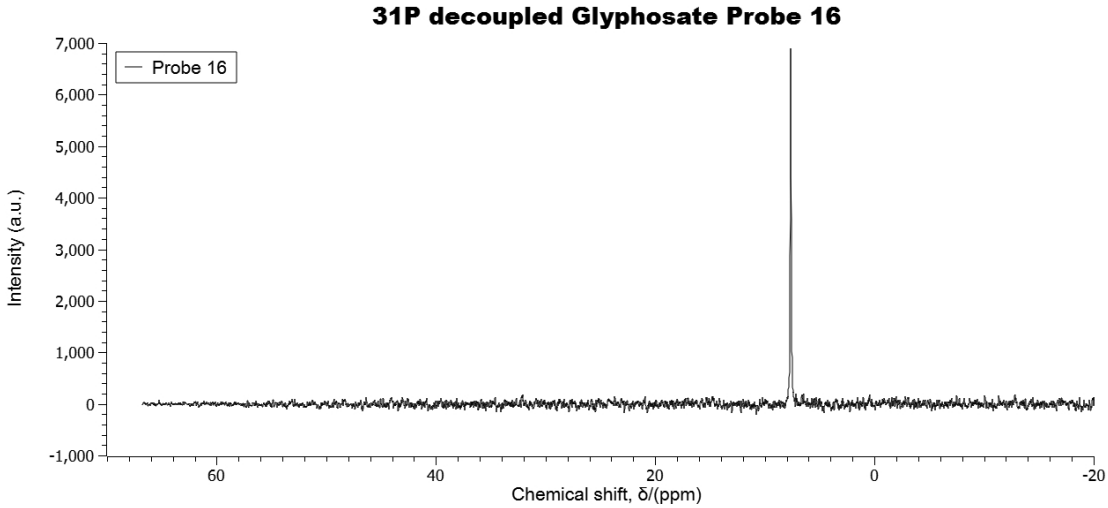

Figure 11: NMR Spectroscopy of 31P decoupled Glyphosate Probe 16 with unknown conc

Table 2: Calibration Table

Probe No.	Concentration (M)	Integral Area
6	0.1438	2258.52
8	0.0479	0819.35
9	0.5636	9469.90
10	0.4697	7678.45
11	0.3757	6020.39
12	0.2818	4617.81
13	0.1879	2927.31
14	0.0939	1444.65
15	0.2348	3625.14
16	Unknown	0801.16

3.6 Calibration

the graphs were plotted using Scidavis software.the integral area is calculated using scidavias software.

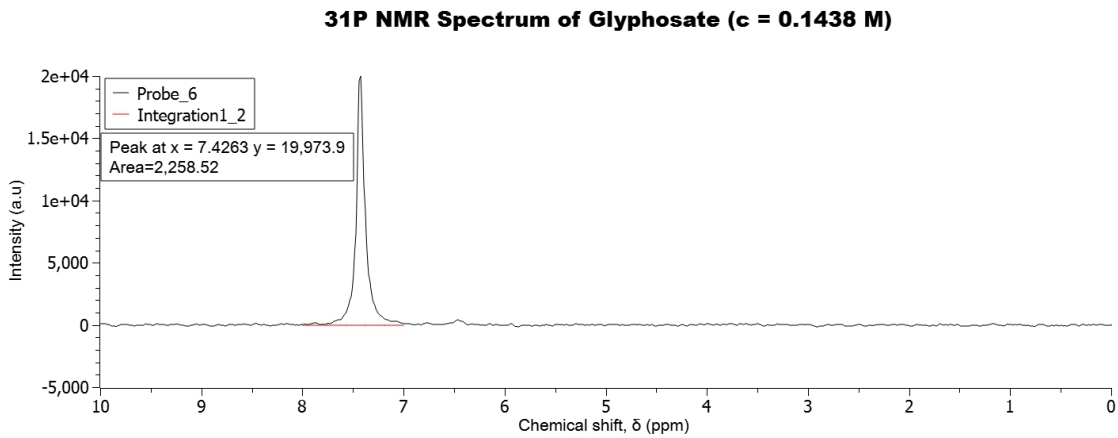

Figure 12: The identification of Area under the graph

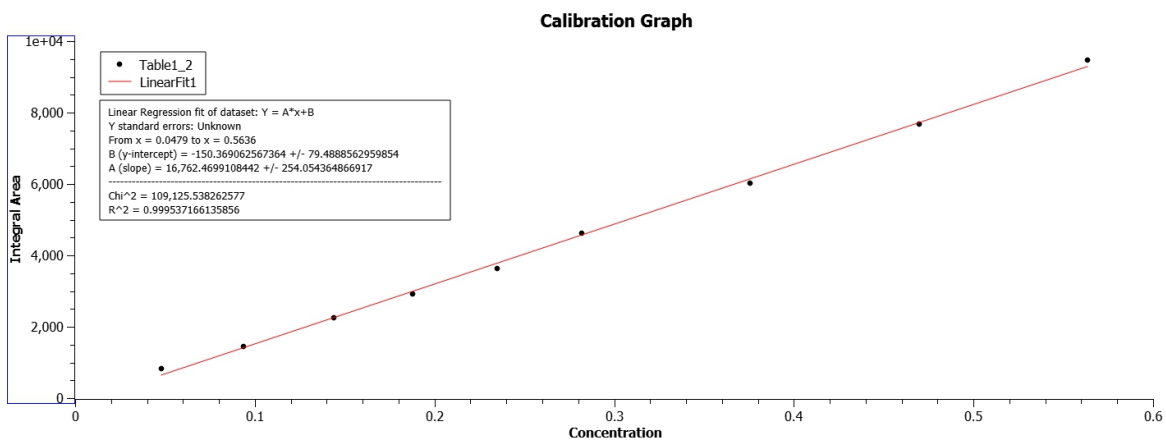

Figure 13: Calibration graph

Determination of Unknown Concentration

In this experiment, the concentration of an unknown sample is calculated using the integral area obtained from NMR measurements. Some samples have known concentrations and corresponding integral areas, which are used as reference.

First, a known standard is selected from the data. Probe number 8 is chosen, which has a known concentration of 0.0479 M and an integral area of 819.35.

The ratio of concentration to integral area is calculated as follows:

$$k = \frac{\text{Concentration}}{\text{Integral Area}} = \frac{0.0479}{819.35} \approx 5.85 \times 10^{-5}$$

This value represents the concentration per unit integral area.

The integral area of the unknown sample is 801.16. The concentration of the unknown sample is calculated by multiplying the ratio with the unknown area:

$$C_{\text{unknown}} = k \times \text{Integral Area} = 5.85 \times 10^{-5} \times 801.16$$

$$C_{\text{unknown}} \approx 0.0468 \text{ M}$$

Therefore, the concentration of the unknown sample is approximately 0.0468 M.

Table 3: Calibration Table

Probe No.	Concentration (M)	Integral Area
6	0.1438	2258.52
8	0.0479	0819.35
9	0.5636	9469.90
10	0.4697	7678.45
11	0.3757	6020.39
12	0.2818	4617.81
13	0.1879	2927.31
14	0.0939	1444.65
15	0.2348	3625.14
16	**0.0468**	0801.16

4 Identification of unknown compound using NMR spectroscopy

To identify the unknown substance, we would perform the CHN analysis.

4.1 CHN Analysis

Elemental analysis was performed to determine the empirical formula of the compound. A 100 g sample was analyzed with the following composition:

Table 4: CHN Analysis Results

Element	Percentage	Mass (in 100g)
Carbon (C)	60%	60 g
Hydrogen (H)	4.48%	4.48 g
Nitrogen (N)	0%	0 g
Oxygen (O)	35.52%	35.52 g
Total	**100%**	**100 g**

4.1.1 Calculation of Number of Moles

The number of moles for each element was calculated by dividing the mass by the respective atomic mass (C = 12.01, H = 1.008, O = 15.999):

Table 5: Number of Moles Calculation

Element	Mass (g)	Atomic Mass	Moles
Carbon (C)	60	12.01	$\frac{60}{12.01} = 4.99$
Hydrogen (H)	4.48	1.008	$\frac{4.48}{1.008} = 4.44$
Oxygen (O)	35.52	15.999	$\frac{35.52}{15.999} = 2.22$

4.1.2 Determination of Mole Ratios

The mole ratios were determined by dividing each value by the lowest number of moles (2.22):

Table 6: Mole Ratio Calculation

Element	Moles	Lowest Moles	Mole Ratio	Whole Number
Carbon (C)	4.99	2.22	$\frac{4.99}{2.22} = 2.25$	9
Hydrogen (H)	4.44	2.22	$\frac{4.44}{2.22} = 2.00$	8
Oxygen (O)	2.22	2.22	$\frac{2.22}{2.22} = 1.00$	4

4.1.3 Empirical Formula and Molar Mass

Based on the mole ratios, the empirical formula is determined to be $\mathrm{C_9H_8O_4}$.

The molar mass of the compound is calculated as follows:

$$
\begin{aligned}
\text{Molar mass} &= (9 \times 12.01) + (8 \times 1.008) + (4 \times 15.999) \\
&= 108.09 + 8.064 + 63.996 \\
&= 180.15 \text{ g/mol}
\end{aligned}
$$

The empirical formula $C_9H_8O_4$ corresponds to acetylsalicylic acid (aspirin), a common organic compound with pharmaceutical applications.

4.2 ^{1}H NMR Spectroscopic Analysis

The ^{1}H NMR spectrum shows four distinct regions that help identify the structure:

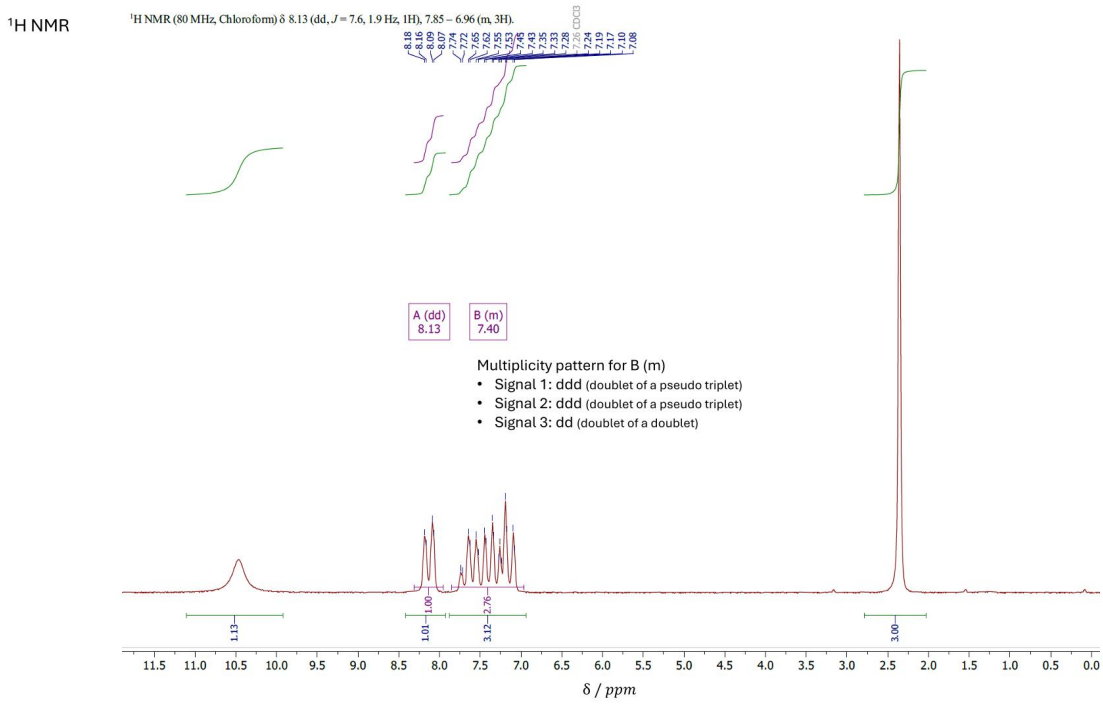

Figure 14: ^{1}H NMR Spectra

4.2.1 Carboxylic Acid Proton

A broad signal is seen at δ 10.5–13 ppm, indicating a carboxylic acid O–H proton. The broadness is due to the rapid proton exchange and hydrogen bonding that acidic protons exhibit.

4.2.2 Aromatic Protons

In the aromatic region (δ 6.5–8.5 ppm), four distinct aromatic protons are observed with complex splitting patterns. Some appear as a doublet of doublets (dd), whereas others appear as more complex multiplets. This pattern shows that all four aromatic protons are magnetically distinct, implying that they are bound to an ortho-disubstituted benzene ring.

4.2.3 Methyl Group

A sharp singlet appears at δ 2.3 ppm when integrating for 3H. This singlet indicates a methyl group next to a carbonyl group. The fact that it is a singlet (not split) indicates that the neighboring carbon does not have any protons attached, implying the presence of an acetyl group.

4.2.4 Absence of Other Groups

There are no signals at δ 9-10 ppm (no aldehyde) or δ 3.3-4.0 ppm (no methoxy groups).

4.3 ^{13}C NMR Spectroscopic Analysis

The ^{13}C NMR spectrum reveals the carbon framework of the molecule:

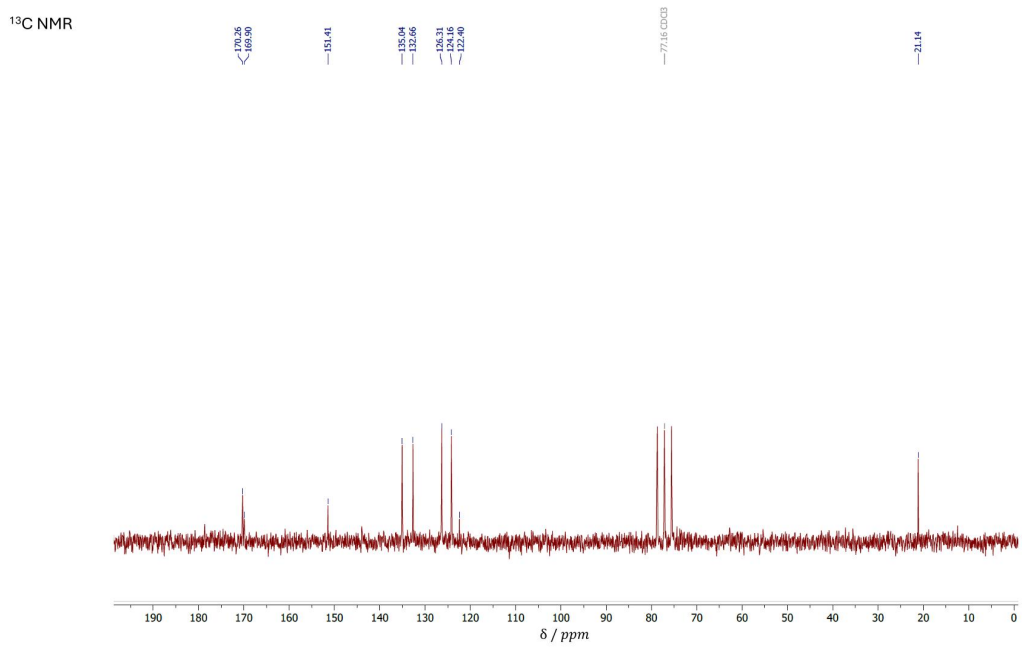

Figure 15: ^{13}C NMR Spectra

4.3.1 Carbonyl Carbons

Two signals appear at δ 165–180 ppm, indicating two different carbonyl groups. One is from a carboxylic acid (–COOH) and the other is from an ester or acetyl group (–O–CO–CH_3). The presence of two separate signals confirms two independent carbonyl functionalities.

4.3.2 Aromatic Carbons

Multiple signals arise between δ 115–140 ppm, representing the six carbons of the benzene ring. The dispersion of signals indicates that each aromatic carbon is in a distinct magnetic environment, indicating an unsymmetrical substitution pattern.

4.3.3 Methyl Carbon

A single signal appears at δ 20–25 ppm, assigned to a methyl carbon bonded to a carbonyl group. This matches the methyl singlet observed in the ^{1}H NMR.

4.4 DEPT Spectroscopic Analysis

DEPT categorizes carbons by the number of attached H (CH3, CH2, CH, or quaternary C).

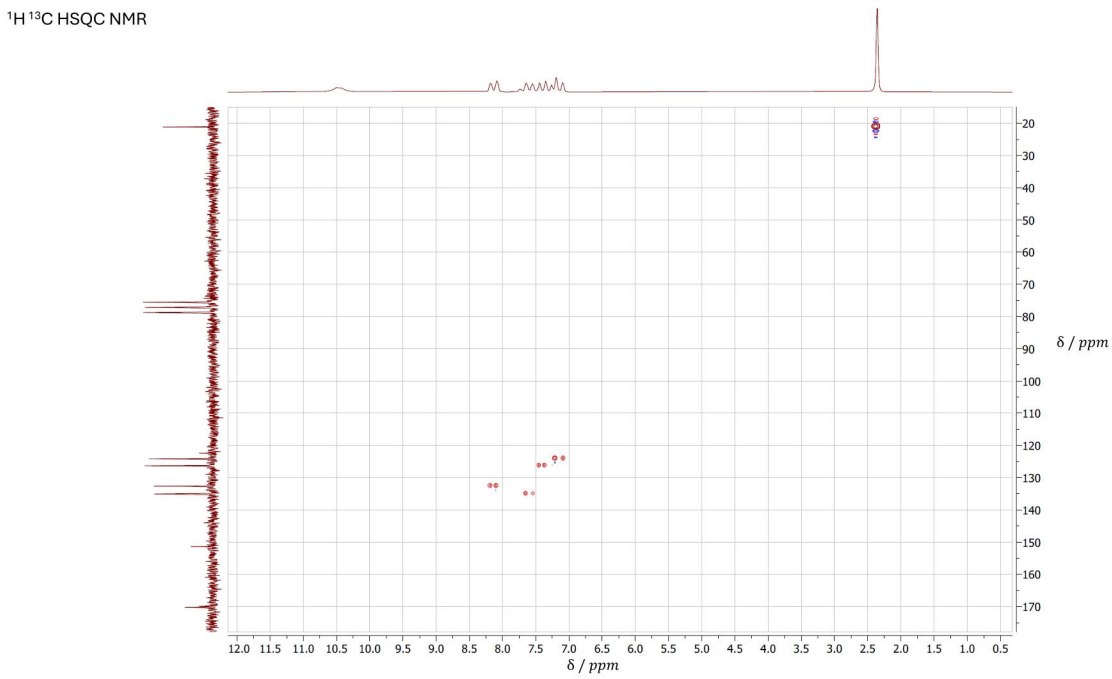

Figure 16: DEPT Spectroscopic graph

4.4.1 Carbonyl Carbons

The two carbonyl signals at δ 165–180 ppm are completely absent in the DEPT spectrum, indicating that they are quaternary carbons with no associated hydrogens. This excludes out aldehydes while confirming carboxylic acid or ester groups.

4.4.2 Aromatic CH Carbons

Several signals remain in the aromatic region (δ 115–140 ppm) in the DEPT, corresponding to aromatic carbons with attached hydrogens (CH). Quaternary aromatic carbons are suppressed.

4.4.3 Methyl Group

The signal at δ 20–25 ppm persists in DEPT-135 with positive phase, confirming it is a methyl (CH_3) group.

4.4.4 No Methylene Groups

Critically, no negative signals appear in the DEPT-135 spectrum, proving the complete absence of CH_2 groups. This rules out longer alkyl chains and confirms a simple methyl group attached to the carbonyl.

4.5 HSQC Correlation Analysis

HSQC shows which protons are attached to which carbons by correlating ^{1}H and ^{13}C signals.

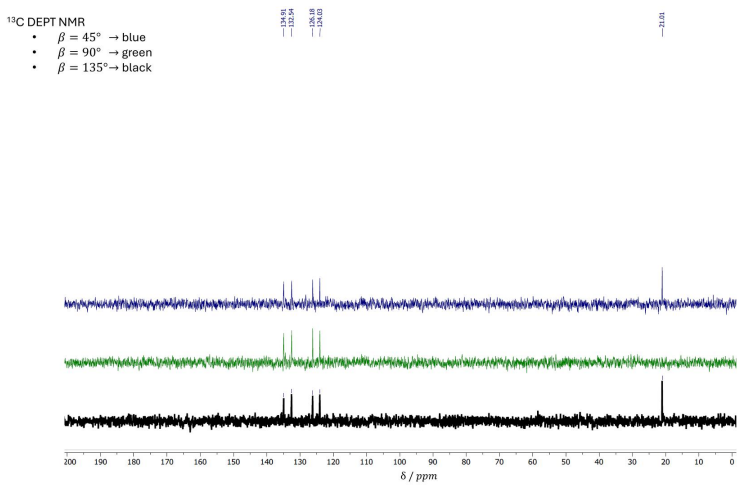

Figure 17: HSQC Correlation graph

4.5.1 Aromatic Region

Each aromatic proton (δ 6.5–8.5 ppm) correlates to one unique aromatic carbon (δ 115–140 ppm), confirming that all four aromatic protons are magnetically distinct.

4.5.2 Methyl Group

The methyl protons at δ 2.3 ppm correlate to the carbon at δ 22 ppm, confirming attachment.

4.5.3 Absence of Expected Correlations

No correlations appear for the carbonyl carbons (quaternary, no protons) or the acidic proton (bonded to oxygen, not carbon). This confirms all C–H bonds are accounted for and validates the structure.

4.6 Structural Determination and Compound Identification

By combining all NMR data with the molecular formula $\mathrm{C_9H_8O_4}$, the structure can be determined with certainty.

4.6.1 Evidence for Ortho-Disubstitution

The complex splitting patterns (dd and ddd) of the aromatic protons indicate an ortho (1,2)-disubstituted benzene ring. A para-substituted (1,4) ring would show a simple symmetrical pattern, which is absent. A meta-substituted (1,3) ring would show some protons appearing as triplets, which is also absent.

4.6.2 Identification of Functional Groups

The broad signal at δ 10.5–13 ppm confirms a carboxylic acid group (–COOH). The methyl singlet at δ 2.3 ppm, combined with its ^{13}C signal at δ 22 ppm, confirms an acetyl group (CH_3–CO–) bonded to oxygen, forming an acetoxy group (–O–CO–CH_3).

This is different from salicylic acid, which would show a phenolic –OH around δ 5–7 ppm with no methyl signal. It is also different from methyl salicylate, which would show a methyl signal at δ 3.9 ppm.

4.6.3 Final Identification

The combination of a carboxylic acid and an acetoxy group in the ortho position on a benzene ring matches perfectly with the molecular formula $C_9H_8O_4$. This compound is **2-acetoxybenzoic acid**, more commonly known as **Acetylsalicylic Acid** or **Aspirin**.

All spectroscopic evidence from CHN analysis, ^{1}H NMR, ^{13}C NMR, DEPT, and HSQC conclusively confirms this identification.

5 Conclusion

This experiment successfully demonstrated the application of NMR spectroscopy to the comprehensive analysis of unknown chemical samples. Elemental analysis (CHN) identified the empirical formula as $C_9H_8O_4$ with a molar mass of 180.15 g/mol. By analyzing ^{1}H NMR, ^{13}C NMR, DEPT, and HSQC data, the unknown compound was identified as **Acetylsalicylic Acid (Aspirin)**. The ^{1}H NMR analysis identified a carboxylic acid proton (δ 10.5–13 ppm), four distinct aromatic protons (δ 6.5–8.5 ppm), and a methyl singlet (δ 2.3 ppm). The ^{13}C NMR revealed two distinct carbonyl carbons, indicating the presence of a carboxylic acid and acetoxy group. DEPT analysis confirmed the absence of methylene groups, whereas the complex splitting patterns of aromatic protons revealed an ortho (1,2) substitution pattern. By systematically excluding para and meta isomers and related compounds such as salicylic acid and methyl salicylate, the structure was conclusively identified as 2-acetoxybenzoic acid (Aspirin, $C_9H_8O_4$). This multi-dimensional NMR approach, combined with elemental analysis, provides a robust methodology for unambiguous structural determination of unknown organic compounds.

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